

# COMMENTS: EM-BASED LIKELIHOOD INFERENCE FOR SOME LIFETIME DISTRIBUTIONS BASED ON LEFT TRUNCATED AND RIGHT CENSORED DATA AND ASSOCIATED MODEL DISCRIMINATION

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The authors are to be commended for their comprehensive, thorough, and insightful investigation of the EM algorithm in estimating the parameters in lifetime distributions from data involving truncation and censoring. They develop detailed steps of the EM algorithm for several widely used distributions based on left truncated and right censored data. These distributions include the lognormal, Weibull, gamma, and generalized gamma distributions. The investigation on the generalized gamma distribution is of important practical value because it contains the exponential, gamma, Weibull and lognormal distributions as special cases, among others. Although the generalized gamma distribution has this nice property, its third parameter  $\kappa$  is usually challenging to estimate, even with censoring data only. Truncation can add a second layer of difficulty to the optimization of the log-likelihood function. The EM algorithm proposed in the paper can provide a more stable solution.

There are some pros and cons to the EM algorithm presented in this paper. A strong benefit is its robustness during the optimization process. When it comes to optimization, one might just simply program the log-likelihood function and throw it in an optimization function provided by some software. Based on our experience, these gradient based algorithms, such as the Newton-Raphson (NR) method, can provide the solution rather quickly if the log-likelihood function has a nice shape and good starting values are provided for the unknown parameters. In most other situations though, these gradient based algorithms can fail and generate errors. Previous experience with the EM algorithm reveals that it tends to be more robust during the optimization process. While it usually takes more iterations to converge, this is not really a problem thanks to modern computing power. Of course, one needs to work out all the detailed steps in the EM algorithm, but they are already available in this paper. The algorithm may also need some extra work to calculate the variance-covariance matrix, perhaps by using the Luis formula.

In regards to the simulation study performed in the paper, it is assuring that the EM algorithm and NR method perform quite similarly. Their study does show that there are biases for the parameter

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estimates, especially for certain distributions like the gamma distribution. This is most likely due to truncation. The simulation study also shows that truncation affects the coverage probability of the confidence interval procedures. One possible remedy is to use bias correction methods. For example, when constructing bootstrap confidence intervals, one can use the bias corrections in Efron and Tibshirani (1993).

The paper uses AIC to select the best model and the simulation study shows that the AIC criterion works very well. There are other alternatives that could be used as well. One option is to use some formal goodness-of-fit tests. However, the correct use of these tests is important. As the comments given in the rejoinder of Meeker and Hong (2014) state, some goodness-of-fit tests are likely to reject all null hypotheses when the sample size is large and tend to accept all null hypotheses when the sample size is small. Another more intuitive way is to do a graphical check. For lifetime data, the probability plot is an effective way of visualizing data (e.g., Chapter 6 of Meeker and Escobar, 1998). It allows one to visually compare the parametric fit with the nonparametric fit. For censored data, the Kaplan-Meier estimator can be used for the nonparametric fit. For data with truncation, some adjustments are required. Page 268 of Meeker and Escobar (1998) describes a method for such adjustments, making it possible to compare the parametric and nonparametric estimates.

The challenge of parameter estimation with left truncated and right censored can sometimes come from the data and not necessarily the model itself. We have had such an experience with a recent dataset. The online supplement of Hong, Meeker and McCalley (2009) describes an estimation problem with lifetime data from a particular group of power transformers, called the MD Group Data. Attempting to fit a Weibull distribution to the MD group data gives nonsensical estimates of the Weibull scale and shape parameters with both maximum likelihood (ML) estimates close to 0. Figure 2 of the online supplement shows that this is caused by the irregular shape of the log-likelihood function. Specifically, the problem occurs due to the enormous spread in the lifetimes (the 5 near 1980, one in 2001, and the remaining 21 that have not failed yet) and the uncertainty due to the truncation of units that failed before 1980. One remedy is to put a prior distribution (i.e., from engineering knowledge) on the scale and shape parameters to restrict the parameter estimates away from nonsensical regions.

Overall, we thank the authors for their contributions on this area and we appreciate the opportunity to offer comments on their work and additional perspectives on the challenges of estimation and inference based on truncated and censored lifetime data.

## References

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